Low-Rank Spectral Fingerprints for AI-Generated Music Detection

Interim report 1: scoping memo

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Domain	Audio forensics / music provenance
Method family	STFT magnitude spectrograms; exact and randomized SVD; feasibility study of low-rank spectral features
LA object	10-s mono clip; STFT magnitude $M \in \mathbb{R}_{\geq 0}^{1025\times 431}$; top $k = 20$ singular values
Algorithms	Exact NumPy/LAPACK SVD versus randomized SVD ($k = 20$, $p = 10$, one power step)
Data	Up to 500 paired clips from FakeMusicCaps and MusicCaps [1]
Scope	Evaluate low-rank spectral features as a binary detection method, not build a universal AI-music detector

1 Problem and aim

The detection of AI-generated music has proven a problem for platforms, rights holders and listeners. In this work, we intend to distinguish AI-generated audio from human-generated audio.

The input is a ten-second audio clip, and the output is a binary prediction: human-generated or AI-generated (text-to-music). Our method extracts the normalized leading singular values of the clip's STFT magnitude spectrogram, the stability of its leading singular subspace over time and features derived from a shared spectral basis, and uses them as features for a logistic regression classifier. The aim is to evaluate whether these low-rank spectral features contain enough information to distinguish the two classes on held-out data, not to create a universal AI-music detector.

Prior work also warns that strong controlled results may not generalize across manipulations or datasets [2, 3], so we try to balance classes and group each original/synthetic caption pair in the same data split.

We will sample up to 500 caption IDs from FakeMusicCaps and retrieve the matching human-made MusicCaps clips. We will choose one generated clip per ID, balance across the five generators where possible, and keep matched clips in the same split. If original clips are inaccessible, we will report the smaller paired sample rather than replace them with unmatched tracks.

Success requires held-out balanced accuracy ≥ 0.65, with the lower bound of its 95% bootstrap confidence interval exceeding 0.50. The approximation must also satisfy $\|\sigma^{\text{rand}} - \sigma^{\text{exact}}\|_2 / \|\sigma^{\text{exact}}\|_2 \leq 10^{-2}$ for the leading 20 singular values of the stacked matrix, checked against exact SVD on a subset; otherwise, randomized SVD will not be considered accurate enough for this feature pipeline.

2 Linear-algebra formulation

Each ten-second clip is converted to mono at 22.05 kHz. A centred short-time Fourier transform (STFT), using a 2048-sample window/FFT and hop 512, gives a magnitude matrix $M \in \mathbb{R}_{\geq 0}^{F\times T}$ with $F = 1025$ and $T \approx 431$. Its singular value decomposition is

$$M = U\Sigma V^\top, \qquad \sigma_1 \geq \sigma_2 \geq \cdots \geq 0.$$

Per-clip matrices are small, so this SVD is computed exactly. We classify the first 20 normalized squared singular values, $p_i = \sigma_i^2 / \|M\|_F^2$, using logistic regression; an entropy-based effective rank will be added and defined precisely in interim report 2. The linear-algebra task is to compute or approximate this leading spectrum, which summarizes how spectrogram energy is distributed across orthogonal components. It may discard useful phase, local-frequency, and temporal cues, so classification is an empirical feasibility test.

The matrix is dense, rectangular and nonnegative, and is not generally symmetric or sparse. Its rank is at most $\min(F, T) = 431$. We exploit its rectangular shape ($F > T$) and the expectation that much of its energy may be concentrated in a relatively small number of singular components.

Subspace stability. We cut M into sliding windows $M^{(j)}$ of w frames with step s, and let $U_j \in \mathbb{R}^{F\times k}$ hold the leading k left singular vectors of $M^{(j)}$. The principal angles $0 \le \theta_1 \le \cdots \le \theta_k \le \pi/2$ between consecutive windows are given by $\cos\theta_i = \sigma_i(U_j^\top U_{j+1})$. Their average over the clip measures how much the dominant spectral subspace changes over time and is used as an additional feature.

Shared spectral basis. Spectrograms of many training clips are stacked into $A = [M_1 \ \cdots \ M_N] \in \mathbb{R}^{F\times 431N}$. Its leading left singular vectors, computed with our randomized SVD, form a spectral basis shared across clips; the basis is used to derive additional per-clip features (to be specified in interim report 2). For A the task is a truncated singular value decomposition: we seek its leading $k = 20$ singular values and vectors without computing the complete decomposition.

3 Approaches and evaluation

Dense thin SVD (per clip). NumPy/LAPACK gives a numerically accurate spectrum; we use it for every clip, and as the reference for randomized SVD. For $F \ge T$, the cost is $O(FT^2 + T^3)$ flops and memory is $O(FT + T^2)$. Its advantage is reliability; its drawback is that it computes the full spectrum, and these moderate matrix sizes may leave little room for speedup.

Randomized SVD (stacked matrix). Following Halko, Martinsson, and Tropp [4], we will apply it to A and implement range finding with target rank $k = 20$, oversampling $p = 10$ ($\ell = k + p = 30$), QR orthogonalization, and one power step ($q = 1$), then take the SVD of the reduced matrix. Its cost is approximately $O((2q+2)FT\ell + (F+T)\ell^2)$ flops and memory $O(FT + (F+T)\ell)$, with T replaced by $431N$, the number of columns of A. It focuses work on the leading subspace, but can lose accuracy when singular values decay slowly or cluster; one power step may then be insufficient, and extra steps increase runtime. We will implement this randomized algorithm and the feature pipeline ourselves, while using NumPy / LAPACK for the dense reference and scikit-learn for logistic regression.

We will compare feature error and wall-clock time, then report balanced accuracy, ROC-AUC and bootstrap confidence intervals on a caption-grouped held-out set. A bandwidth-only classifier is a diagnostic baseline: if it performs well, labels may be explained by codec or spectral-bandwidth artifacts rather than music structure. Frobenius normalization reduces sensitivity to overall gain, but codec and bandwidth differences can still leak into labels; identical mono conversion, resampling and loudness normalization will be applied to both classes.

4 Risks and work plan

The main risks are unavailable matched originals, a flat singular-value spectrum, and the generator or mastering artifacts skewing the classifier. If the target accuracy is missed, we will report this negative result.

Chosen methods. We will implement randomized SVD with QR orthogonalization, oversampling and one power iteration, since it directly targets the leading singular subspace and allows us to investigate the computational advantages of approximate decomposition. NumPy/LAPACK SVD will serve as the numerical reference. We will implement the randomized range-finding procedure, feature extraction and numerical error evaluation ourselves, while using NumPy/LAPACK for the reference decomposition and scikit-learn for logistic regression.

Work plan. By 9 November, we will verify the availability of matched clips and finalize the preprocessing and caption-grouped split protocol.

By 23 November (interim report 2), we will complete the paired dataset preparation and preprocessing pipeline, implement randomized SVD and normalized singular-value feature extraction, validate approximation error and runtime against dense SVD on a representative subset and run the initial logistic regression and bandwidth-only baselines.

By 14 December (final report), we will: Complete the evaluation on up to 500 matched caption IDs, Compare exact and randomized SVD in approximation error and runtime, Evaluate classification performance using balanced accuracy, ROC-AUC and bootstrap confidence intervals, Analyse potential dataset, codec, bandwidth and generator-related biases and discuss limitations, and Complete the final report and present the results

Division of work. Iryna will handle dataset acquisition, matching, preprocessing and split construction. Petro will implement the SVD, feature extraction and numerical validation against dense SVD. Both will contribute to classification evaluation, interpretation of results and report writing.

References

[1] L. Comanducci, P. Bestagini and S. Tubaro, "FakeMusicCaps: A Dataset for Detection and Attribution of Synthetic Music Generated via Text-to-Music Models," *Journal of Imaging*, vol. 11, no. 7, article 242, 2025. doi:10.3390/jimaging11070242.

[2] D. Afchar, G. Meseguer-Brocal and R. Hennequin, "AI-Generated Music Detection and its Challenges," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025. doi:10.1109/ICASSP49660.2025.10890655.

[3] Y. Li, Q. Sun, H. Li, L. Specia and B. W. Schuller, "Explainable detection of machine generated music and early systematic evaluation," *Scientific Reports*, vol. 16, article 13757, 2026. doi:10.1038/s41598-026-42133-7.

[4] N. Halko, P. G. Martinsson and J. A. Tropp, "Finding Structure with Randomness: Probabilistic Algorithms for Constructing Approximate Matrix Decompositions," *SIAM Review*, vol. 53, no. 2, pp. 217–288, 2011. doi:10.1137/090771806.